Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Test 3. Mutual information and capacity (key)

1. A random symbol X takes values in a finite set of at least two elements. Its entropy $H(X)$ is largest when

(a) the values are equiprobable. ⋆
(b) one value has probability one.

Solution: The answer is (a). Section 3.1 gives $H(X) \leq \log_2 |\mathcal{X}|$, with equality for equiprobable values; one value of probability one gives $H(X) = 0$.

2. On the bit-flipping channel with $p = \frac{1}{2}$ and equiprobable inputs, the output is an equiprobable bit, so $H(Y) = 1$. The mutual information $I(X;Y)$ is

(a) 1 bit per channel use.
(b) 0. ⋆

Solution: The answer is (b). By Proposition 3.4, $I(X;Y) = 1 - h_2(\frac{1}{2}) = 0$: the uncertainty of the output is all noise, as checked in lecture 5.

3. For a fixed memoryless channel with a finite input alphabet, its capacity is the largest mutual information obtained by varying

(a) the receiver's decision rule.
(b) the input distribution. ⋆

Solution: The answer is (b). Definition 3.7 gives $C = \max_{\Pr(x)} I(X;Y)$: the input probabilities vary, the channel law is fixed, as in Seminar 5.

4. For a memoryless channel of capacity C and a fixed rate $R < C$, the coding theorem says that

(a) there are codes of rate R, long enough, with error probability as small as we wish. ⋆
(b) every code of rate R has error probability as small as we wish if it is long enough.

Solution: The answer is (a). Theorem 3.9 is an existence result: good codes exist at every rate below capacity, at the price of their length.

5. Let $Y = X + Z$, where Z has a density and is independent of X. Assume the differential entropies below are finite. Changing the input distribution

(a) can change both $h(Y)$ and $h(Y|X)$.
(b) can change $h(Y)$, but leaves $h(Y|X) = h(Z)$ unchanged. ⋆

Solution: The answer is (b). Section 3.4 gives $h(Y|X) = h(Z)$ for every input law, so by (3.24) only $h(Y)$ depends on the input, as in Seminar 5.

6. For discrete random variables X and Y, the mutual information $I(X;Y)$ measures

(a) the average reduction in uncertainty about X when Y is observed. ⋆
(b) the average uncertainty about X that remains after Y is observed.

Solution: The answer is (a). Section 3.2 gives $I(X;Y) = H(X) - H(X|Y)$, the uncertainty removed by observing Y. The remaining uncertainty is $H(X|Y)$.

7. The differential entropy of a real random variable with a density

(a) is always nonnegative.
(b) can be negative. ⋆

Solution: The answer is (b). Example 3.11 gives $h(U) = \log_2 a$ for a uniform variable on an interval of length a, negative when $a < 1$.

8. A real Gaussian channel has noise of variance σ^2 and input constraint $\mathrm{E}(X^2) \leq E_\mathrm{s}$. With $\mathrm{SNR} = E_\mathrm{s}/\sigma^2$, its capacity in bits per channel use is

(a) $\frac{1}{2}\log_2(1 + \mathrm{SNR})$. ⋆
(b) $\log_2(1 + \mathrm{SNR})$.

Solution: The answer is (a). Section 3.5 gives $\frac{1}{2}\log_2(1 + \mathrm{SNR})$ for the real channel; the other option is the formula of the complex channel.

9. A channel input takes M values. Whatever the channel and the input probabilities, $I(X;Y)$

(a) is at most $\log_2 M$, since $I(X;Y) \leq H(X)$. ⋆
(b) can exceed $\log_2 M$ if the noise is small enough.

Solution: The answer is (a). Sections 3.2 and 3.1 give $I(X;Y) \leq H(X) \leq \log_2 M$; in Seminar 5, two inputs reach this ceiling of 1 bit once $d \geq 1$.

10. Among real random variables with a density and a given positive variance, the largest differential entropy is attained by

(a) a uniform distribution.
(b) a Gaussian distribution. ⋆

Solution: The answer is (b). Theorem 3.13 states that the Gaussian maximizes differential entropy at a given variance. The uniform distribution maximizes it under a fixed bounded-support constraint instead.